



FLEET & SHORE PERSONNEL TRAINING

Fatigue Management

Recognising fatigue, understanding its causes, and managing the risk on board.



IMO MSC.1/Circ.1598

MLC 2006

STCW VIII/1

Company SMS

What we will cover

01

Understanding Fatigue

What it is and what causes it

02

The Five Causal Factors

Seafarer, management, ship, environment, operation

03

Fatigue and the Company

Duty of care, scheduling and resources

04

Fatigue and the Seafarer

Signs, symptoms and personal responsibility

05

Hours of Work and Rest

Limits, records and verification

06

Awareness and Training

Objectives and lessons learned



SECTION 01

Understanding Fatigue

Fatigue is a hazard — not an unavoidable part of the job. It must be identified and managed like any other risk.

01

CORRECTING A COMMON BELIEF

Fatigue does not "come with the job"

Fatigue is a hazard that affects safety, health and well-being. It presents a considerable risk to safety of life, property, health, security and protection of the marine environment — and it must be addressed, not accepted.

It is a hazard

It is manageable

It is everyone's business

Why the maritime industry is especially exposed

Fatigue is a problem for every 24-hour transport mode. Shipping adds operational complexity that other industries do not face.



Variety of ship types

Each trade, manning arrangement and machinery configuration produces a different workload pattern.



Pattern and length of sea passage

Short passages give little chance to recover; long passages bring monotony and disturbed routine.



Number of port visits and rotations

Frequent calls mean repeated peaks of intense activity with reduced sleep opportunity.



Time the ship remains in port

Short stays compress cargo work, inspections, bunkering and stores into the same window.



Layered external demands

Inspections, surveys, audits, visits, reporting and security measures all land on the same crew.



The combination is the problem

No single factor causes fatigue on its own. It is the unique combination of schedule, environment, workload and personal condition that produces it — which is why the assessment has to look at all five factor groups.

What actually causes fatigue

Fatigue is caused by a range of factors, but primarily by these six.



Lack of sleep

Inadequate restorative sleep — the single largest contributor.



Poor quality sleep and rest

Sleep that is broken, shallow or repeatedly interrupted does not restore.



Work at the wrong body time

Working or sleeping against the body clock (circadian rhythm).



Long time awake

Staying awake for extended periods degrades performance measurably.



Stress

Psychological load consumes the same reserves as physical work.

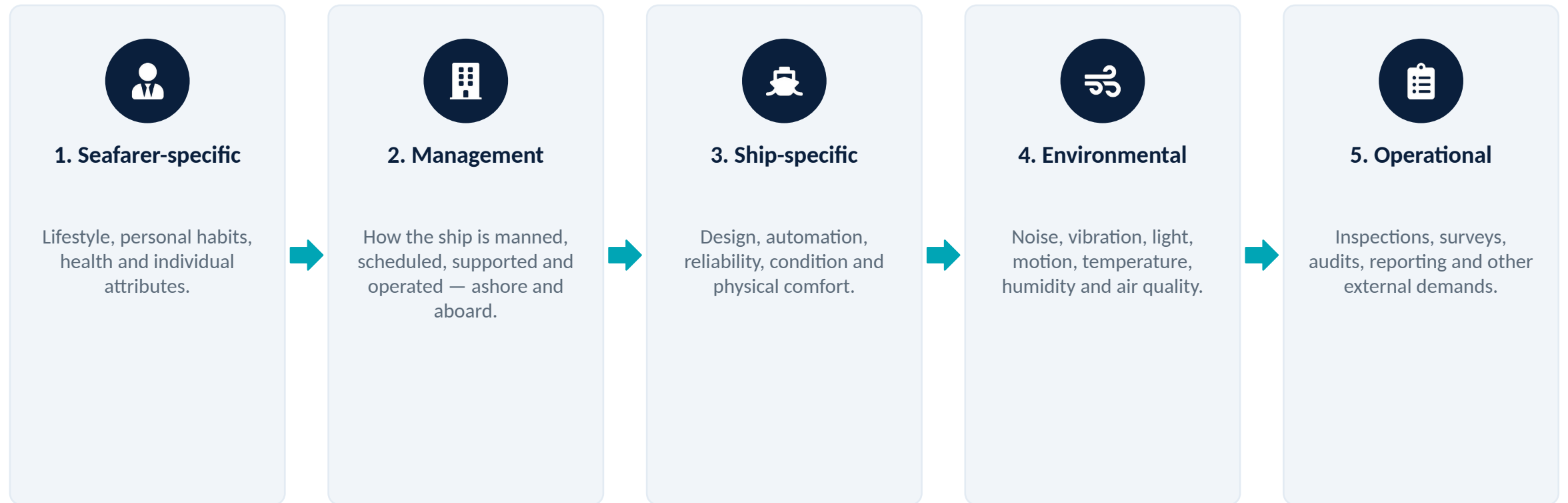


Excessive workload

Prolonged mental and/or physical exertion without adequate recovery.

Five general factor groups

To cover the causes thoroughly, they are grouped into five categories. Every fatigue assessment should consider all five.





SECTION 02

The Five Causal Factors

Fatigue varies from person to person, and its effects depend on the activity being performed.

Seafarer-specific factors

Related to lifestyle behaviour, personal habits and individual attributes.



Sleep and rest

Quantity, quality and continuity of sleep • sleep disorders and disturbances • recovery rest and breaks.



Body clock

Circadian rhythm — the daily cycle that determines when the body is naturally alert or sleepy.



Psychological and emotional

Fear • monotony and boredom • loneliness and social isolation.



Health and well-being

Diet, nutrition and hydration • exercise and fitness • illness and the onset of illness.



Schedule and workload

Age • shift work and work schedules • mental and physical workload • jet lag after travel.



Medication and substances

Alcohol • prescription and non-prescription drugs • supplements • caffeine and other stimulants.

Management factors — organisational

How ships are managed and operated. These factors can cause stress and increased workload, ultimately resulting in fatigue.

- Manning policies, manning levels and crew retention
- Role of riding squads and shore personnel on board
- Administrative work, reporting and inspection requirements
- Economics and commercial pressure on the operation
- Duty schedule — shifts, overtime and breaks
- Company procedures, culture and management style
- Shore-based support actually available to the ship
- Rules, regulations and other resources provided
- Maintenance and repair burden carried by the crew
- Drill schedules and the training programme for the crew



A management issue first

Most of these are beyond the influence of any individual seafarer. That is why fatigue is treated as a shared responsibility — the Company creates the conditions, the seafarer manages within them.

Management factors — voyage and scheduling



Frequency and duration of port calls

Short, frequent calls concentrate cargo work, inspections and stores into the smallest possible window.



Time between ports

The recovery opportunity available before the next peak of activity.



Routing

The chosen route determines exposure to traffic, weather and navigational workload.



Weather and sea condition on route

Heavy weather adds motion-induced fatigue on top of the normal workload.



Traffic density on route

Dense traffic removes any possibility of reduced bridge attention.



Nature of duties and shore leave

Workload in port and at sea, and whether shore leave is actually available.

Ship-specific factors

Some design features affect workload, some affect the crew's ability to sleep, and others affect physical stress on the crew.



Ship design

General arrangement, working spaces and access routes.



Automation

Level and complexity of automation, and level of redundancy.



Equipment

Equipment design and reliability; inspection and maintenance burden.



Condition of the ship

A poorly maintained ship transfers its problems onto the crew.



Work spaces

Physical comfort in the spaces where work is actually performed.



Location of quarters

Proximity to machinery, alarms, mooring decks and traffic routes.



Ship motion

Motion characteristics of the hull in the prevailing sea state.



Accommodation comfort

Physical comfort of accommodation spaces — where recovery must happen.

Environmental factors

Conditions where seafarers live and work, inside and outside the ship, affect both sleep quantity and quality. Long-term exposure can also damage health.



Noise

Main engines, switchboards, TV and conversation affect the ability to fall asleep, cause sleep loss, or alter sleep stage and depth.



Vibration

Changes in vibration pattern can keep people awake, prevent them reaching deeper sleep, or wake them up.



Light

Colour, intensity and timing all matter. Blue light from screens, televisions and smartphones influences the body clock and delays sleep onset — especially before bedtime.



Ship motion

May interfere with sleep, cause seasickness, and produce motion-induced fatigue from the extra energy spent maintaining balance.



Temperature and humidity

Excessive heat or cold reduces alertness. The body sleeps best between 18 °C and 24 °C, so cabin conditions must be controllable.



Ventilation and air quality

Beyond temperature and humidity, noxious odours and stale air degrade both sleep and alertness.

The most significant aspects of fatigue

These eight concepts underpin everything that follows in this course.



Sleep

Quantity, quality and continuity.



Body clock

The circadian rhythm and its daily cycle.



Time awake

Performance falls with hours since waking.



Jet lag

Body clock displaced by long-distance travel.



Workload

Mental and physical demand over time.



Stress

Draws on the same reserves as physical effort.



Health

Nutrition, hydration, fitness and illness.



Individual differences

The same schedule affects two people differently.



SECTION 03

Fatigue and the Company

Research has established a clear link between fatigue and accidents at sea. Addressing it improves safety and reduces cost.

03

What fatigue does to people and to performance



Health effects

Disrupted circadian rhythms, digestive problems, confusion, lethargy, respiratory problems, depression and irritability. Long-term sleep loss is linked to cardiovascular and gastrointestinal disease, mental health problems and chronic stress.



Performance effects

Diminished attentiveness and concentration, slowed physical and mental reflexes, and impaired rational decision-making. Fatigue greatly increases the risk of incidents and injuries in the workplace.



Commercial effect

Fewer injuries and less physical damage to high-value assets and to the environment. Managing fatigue has the potential to cut cost, not add it.



It affects everyone

Fatigue affects all individuals regardless of skill, rank, knowledge or training. Seniority provides no protection.

How the Company ensures fatigue prevention on board

- ISM Code requirements for clear, concise guidance on operational procedures on board
- Adequate resources, including manning levels appropriate to the trade
- A safety reporting culture with open communication and no fear of reprisal
- Joining seafarers adequately rested before assuming duties
- Scheduled time for a proper handover at crew change
- Realistic voyage length, time in port, length of service and leave ratios
- Attention to multicultural issues, language barriers and social, cultural and religious isolation
- Management of interpersonal relationships, stress, loneliness, boredom and social deprivation
- Provision for shore leave, onboard recreation and family communication
- Sound watchkeeping arrangements and job rotation where practicable
- Adequate sleeping berths and accommodation; adequate quality and quantity of food
- Modification of present or future ship design where necessary



Training reaches ashore too

- Shore staff whose decisions affect fatigue must be trained
- This includes resource planning and manning levels
- And duty scheduling decisions
- Their decisions affect seafarer fatigue and shipboard safety

Seven duty schedule considerations

The Company is responsible for ensuring duty schedules provide adequate opportunity for sleep, and must at minimum comply with STCW regulation VIII/1.



Work hours

The length and placement of work periods.



Rest hours

Rest periods between work periods.



Night watches

Night watches or work against the body clock.



Short breaks

Short rest breaks within work periods.



Naps

Planned naps where a policy permits them.



Recovery sleep

Sleep that repays an accumulated deficit.



Reset breaks

Longer breaks that reset the body clock.



Sleep inertia

Allow time to become alert before critical tasks.



SECTION 04

Fatigue and the Seafarer

Fatigued individuals are poor judges of their own fatigue — because fatigue impairs exactly the judgement needed to assess it.

04

Who is responsible for what



The Company

Creates the conditions

- Provides a work and living environment that minimises fatigue-related risk
- Sets manning levels and duty schedules that allow adequate sleep
- Provides shore-based support and adequate resources
- Delivers training to sea staff and to shore staff whose decisions matter



The Seafarer

Manages within them

- Uses the time available for rest and sleep appropriately
- Ensures their own behaviour does not create or increase risk
- Monitors and manages hours of sleep
- Reports fatigue-related hazards that affect safety

Cognitive signs and symptoms

Signs are grouped as cognitive, physical and behavioural. You will usually notice them in others before you notice them in yourself.



Fixates on the trivial

Focuses on a minor problem while neglecting more important ones.



Slow or no response

Delayed reaction to normal, abnormal or emergency situations.



Lapses of attention

Short gaps where nothing is registered at all.



Poor judgement

Misjudges distance, speed, closing rate and time.



Forgets tasks

Forgets to complete a task, or part of a task.



Cannot think clearly

Difficulty concentrating and reasoning through a problem.

Physical and behavioural signs



Physical

- Inability to stay awake — head nodding, falling asleep involuntarily
- Difficulty with hand-eye coordination, such as switch selection
- Speech difficulties — slurred, slowed or garbled
- Increased frequency of dropping objects such as tools or parts
- Digestion problems



Behavioural

- Decreased tolerance and/or anti-social behaviour
- Irregular or atypical mood changes — irritability, tiredness, depression
- Ignores normal checks and/or procedures
- Increasing omissions, mistakes and carelessness



Long-term effects

- Cardiovascular disease
- Gastrointestinal disease
- Mental health problems
- Chronic stress
- Reduced fitness for duty over a career

Seafarer responsibilities in fatigue risk management

Many mitigation measures — voyage scheduling, ship design, work scheduling — are beyond any one person. These are the ones that are not.



Report for duty fit to work

Commence the duty schedule in a fit state to work the expected duty length and perform assigned work safely.



Manage your sleep

Monitor and effectively manage hours of sleep, and use available rest periods appropriately.



Report fatigue hazards

Report fatigue-related hazards that affect safety, through the ship's reporting system.



Communicate about safety

Maintain appropriate communication about safety with the watch, the department and the Master.



Know how to counter it

Be aware of fatigue and of the personal strategies that reduce its effects.



Look after your health

Monitor and seek treatment for health concerns that may affect fitness for duty — including sleep difficulties.



SECTION 05

Hours of Work and Rest

The limits, the records, and who is accountable for each of them.

05

Responsibilities on board and ashore



The Master shall

Ensure compliance with these procedures ·
Ensure deviations from statutory work/rest requirements are reported to the Company by e-mail to the DPA · Record and monitor his own work/rest hours.



Chief Engineer and Chief Officer shall

Ensure every seafarer in their department records work/rest hours · Be directly responsible for planning and monitoring those periods · Report any deviations to the Master · Record and monitor their own hours.



The Company shall

Monitor rest-hour deviation reports and act — stopping operations to rest the crew, waiting at anchor or lay-by, providing additional manning · Ensure adequate resources and shore-based support to comply with the rules in force.

The numbers every seafarer must know

Work and rest periods for all seafarers shall comply with the requirements of ILO / MLC 2006.

14

Maximum hours of work in any
24-hour period

72

Maximum hours of work in any
7-day period

10

Minimum hours of rest in any 24-
hour period

77

Minimum hours of rest in any 7-
day period



The 24-hour period — Begins when the seafarer starts work immediately after a period of at least 6 consecutive hours off duty.



Splitting rest — Rest may be divided into no more than two periods, one of which must be at least 6 hours; the interval between consecutive rest periods must not exceed 14 hours.



Short breaks — Any break of less than 30 minutes is not counted as a period of rest.

Watch systems and additional manning

- The Company prohibits six-on / six-off watches except in an emergency, and requires a three-watch system on board.
- A two-watch system such as 6 on / 6 off may appear compliant in the short term, but tends to create technical non-conformances in each rest period under both STCW and MLC.
- Any period of additional work on a two-watch system is likely to produce significant non-conformances in subsequent work periods.
- It is Company policy to provide additional manning, particularly where voyages are short or workloads are high — including STS operations.
- This policy is verified by checking the manning arrangements of vessels on short voyages with high workloads.
- If additional manning cannot be obtained, anchoring the vessel should be considered in order to avoid fatigue.
- Joining seafarers who have travelled more than about 12 hours should take up duties the following day, to allow adequate rest and effective handover.
- No more than 36 hours of work in any 72-hour period where the OPA 90 requirement applies for USA calls (ISF monitor).



When it cannot be met

If the requirement cannot be implemented, the Master informs the Company. The Master has authority to take appropriate measures — waiting at anchor, suspending operations, re-assigning duties, or requiring additional manning where vessel capacity permits.

The Master's authority in an emergency

Nothing in these rules impairs the Master's right to require any hours of work necessary for the immediate safety of the ship, persons on board or cargo, or to assist other ships or persons in distress.



Suspend the schedule

The Master may suspend the schedule of work or rest hours and require the necessary work until the normal situation has been restored.



Compensate afterwards

As soon as practicable after normal conditions return, seafarers who worked during a scheduled rest period must be given an adequate period of rest.



Record and notify

The event is recorded in the log books and notified to the Company. STCW requirements covering overriding operational conditions apply.



Drills

Emergency drills and drills required by national or international instruments are conducted so as to minimise disturbance of rest periods and not induce fatigue.



UMS ships

Where continuous engine-room watchkeeping is not carried out, periods of responsibility are allocated to minimise disturbance of rest; adequate compensatory rest is given if rest is disturbed by call-outs.

Monitoring of work and rest hours



Monthly signed records

A record of work/rest hours for all shipboard personnel is maintained in English, signed by the seafarer and the Master monthly and before signing off. One copy is retained on board; one copy is kept by the seafarer.



Posted schedule

The Company posts, in an accessible place on board, a table of the schedule of service at sea and in port and the minimum hours of rest for each position — in the ship's working language and in English.



Non-conformances go ashore immediately

Records of non-conformances are sent to the Company at once for monitoring and action. A monthly summary is sent even if the pages are blank.



Trends are monitored ashore

The Company monitors trends and takes action to reduce or remove recurring non-conformances. Records are reviewed by the responsible superintendent.



Daily review on board

Rest-hour records should be reviewed at least daily by the Master or a person authorised by him.

Records, watch lists and verification

- For every position, the watch list shall contain at least the schedule of rest and work periods at sea and in port, and the maximum hours of work or minimum hours of rest.
- The table of working arrangements shall be established in the appended format in the software programme, maintained on record, and displayed at a prominent location on board.
- The Company-provided software shall be used where installed. If it is inoperative or unavailable for any reason, manual records shall be maintained in the format provided.
- Verification of this procedure is carried out during Superintendent's inspections, reviews and audits, and cross-checked against records such as the port log to confirm accuracy.
- All related records must be kept available on board for inspection for the last two years.
- All related records, posters and tables must be written in English and in the working language of the ship.



When safety may be compromised

Where the work/rest cycle of key personnel has deviated from the regulations and the vessel's safety may be compromised, the Master shall suspend the intended operation until those personnel are adequately rested — keeping the suspension as short as possible — and shall contact the Company immediately.



SECTION 06

Awareness and Training

Fatigue management has to be taught in a way that seafarers can relate to personally — because they will have to make operational decisions based on it.

06

What this training is meant to achieve



Awareness and acceptance

That everyone experiences fatigue — it is not a personal shortcoming or a weakness. Removing the stigma is what makes honest reporting possible.



Know the signs

Short- and long-term signs and symptoms of fatigue, its effects, and the preventive and mitigating measures available.



Build strategies

The ability to develop and implement fatigue management strategies that prevent or minimise fatigue on board.

Tailoring the training, and learning from experience



Who must be trained

- All personnel who work on ships
- Shore-based personnel who contribute to fatigue management
- Those involved in operational risk assessment and resource allocation
- Those who set manning levels and duty schedules
- Content adapted to the knowledge and skills each group needs



What every group learns

- The dynamics of sleep loss and recovery
- The effects of the daily cycle of the body clock
- The influence of workload
- How these factors interact with operational demands to produce fatigue
- How to manage personal fatigue and sleep issues



Learning from experience

- Review shared personal experiences and draw out the lessons
- Focus on relevant case studies from the seafarer's own workplace
- Trainees bring their own perceptions of fatigue and how to mitigate it
- Build a common understanding of the issue and its management
- Put the knowledge into practice at the workplace



Rested crews make better decisions

Fatigue is a hazard we can measure, plan for and report. Managing it is a shared duty between the Company and every seafarer on board.



Protect your sleep opportunity



Watch for the signs in others



Report deviations, never hide them



Record hours honestly and daily